Rohith Sudarshan Anand

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EDUCATION

Georgia Institute of Technology **2025 – 2027**
M.S., Electrical & Computer Engineering *Atlanta, GA*
- Coursework: CMOS Analog Integrated Circuits, Data Converter Design, Sensor & Interface IC Design, Microwave Engineering

University of Illinois at Urbana Champaign **2021 – 2025**
B.S., Electrical Engineering (3.6 GPA) *Champaign-Urbana, IL*
- Coursework: RF & mmWave IC Design, Semiconductor Device Physics, Digital VLSI, RF & Microwave Measurements, Python

SKILLS

- **Design:** ADCs, OTAs/Op-Amps, DACs, gm-C Integrators, LNAs, Mixers, VCOs, Bias Circuits, RF Front-Ends.
- **Testing:** VNA, Spectrum Analyzer, Signal Generator, Oscilloscope, Analog /Mixed-Signal Measurement & Automation.
- **Software:** Cadence Virtuoso, Spectre, Verilog-A, MATLAB, Simulink, Keysight ADS, EMX, Python,

ANALOG / MIXED SIGNAL DESIGN EXPERIENCE

GAMMA Lab (Dr. Shaolan Li) **Aug 2025 – Present**
Graduate Student Researcher (CT-ΔΣ, Pipeline & SAR ADC Design) *Atlanta, GA*
- Model ADC behavior in MATLAB/Simulink, analyzing NTF/STF, noise, and architecture tradeoffs before circuit design
- Implement gm-C integrators, OTAs, and DACs in Cadence to develop low-power ADC blocks in 45 nm CMOS.
- Research multi-stage Miller and feedforward OTA compensation in Cadence, optimizing GBW, phase margin, noise, and power.

A 300 MSPS 73 dB SNDR Pipelined ADC in 45 nm CMOS **Aug 2025 – Dec 2025**
- Developed > 73 dB SNDR pipelined ADC with seven 3.5b/1.5b/4b flash stages consuming < 80 mW power from a 1.2V supply.
- Modeled CDAC mismatch, comparator offset, and kT/C noise in MATLAB to guide pipeline architecture and noise-budget tradeoffs.
- Designed and simulated 2-Stage miller-compensated OTA achieving 73 dB DC gain with 1 GHz BW consuming under 4 mW.

IIT Madras Integrated Circuits Lab (Dr. Nagendra Krishnapura) **June 2025 – Aug 2025**
Analog / Mixed Signal Design Summer Research Intern *Chennai, TN*
- Analyzed a 10-bit SAR ADC with monotonic charge-redistribution DAC to meet <500 µW power target at 1.2 V in TSMC 65 nm.
- Performed Cadence Virtuoso transient simulations to evaluate SNR, ENOB, and THD across PVT corners.

1.8 V to 1.2 V Low-Dropout Voltage Regulator (LDO) in 180 nm CMOS **Jan 2024 – May 2024**
- Designed a telescopic-cascode error-amplifier LDO regulating 1.8 V to 1.2 V across 0.1–10 mA loads, achieving 72 dB amplifier gain, 80–86 dB loop gain, and 18°–75° phase margin with ~6.5 µA bias current.
- Verified AC/DC loop response and PSRR, meeting ≤500 µV/V line- and ≤50 µV/mA load-regulation targets across load conditions.

PRIOR RFIC DESIGN EXPERIENCE

Skyworks Solutions **Jan 2026 – May 2026**
RFIC Design Co-Op *San Jose, CA*
- Contributed to RFIC development of LNAs, couplers & active inductors for multi-band 5G RF front end modules in tsmc40n RF-SOI.
- Designed EM-optimized couplers to achieve >25 dB directivity across 3.3–5 GHz 5G UHB sub-bands and antenna configurations.
- Developed SpectreRF testbenches with HFSS .sNp S-parameter models for pre-silicon EM co-sim and PVT/corner sweeps.

Integrated Circuits & Systems Group (Dr. Mostafa G. Ahmed) **Aug 2024 – May 2025**
RFIC Design Student Researcher (Senior Thesis) *Champaign-Urbana, IL*
- Researched emerging standards and receiver requirements for 6G FR3 Band (7-24 GHz) using IEEE & 3GPP publications.
- Investigated Wideband Receiver Architectures for high-speed wireless communication, optimizing for noise, linearity and gain.
- Designed a 7-24 GHz LNA with wideband Chebyshev input network in a 45 nm CMOS process that achieves 13 dB gain, NF < 6 dB, and 5 dBm OIP3 while drawing < 4 mA current from a 1.2 V source.

5 GHz RF Receiver Front End (ECE 598 Adv RF/mmWave IC Design) **Jan 2024 – May 2024**
- Developed a Hartley Image Rejection Receiver in 45nm CMOS achieving 34 dB gain, S11 < –10 dB, and OIP3 of 13.3 dBm.
- Designed a 3–10 GHz Wideband LNA with 16 dB gain, NF < 4 dB, 10 dBm OIP3, drawing < 3 mA current from 1.2 V supply.
- Designed a 5.2 GHz VCO with phase noise < –120 dBc/Hz and polyphase filters for quadrature LO generation.
- Implemented double-balanced passive mixer with –7 dB conversion gain and image rejection ratio of 52 dB at 200 MHz IF.